

UNION TOOL

Tungsten Carbide End Mills UNIMAX Series

HARDMAX COAT 4 Flute Square End Mills

HMES SP

New

Total 15 models

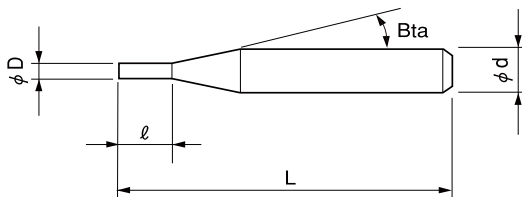


UNION TOOL CO.

Size $\phi 1 \sim \phi 12$ **HMES SP**

Material Applications (☆ Highly Recommended ◎ Recommended ○ Suggested)

Work Material															
CARBON STEELS S45C S55C	ALLOY STEELS SK/SCM SUS	PREHARDENED STEELS NAK HPM	HARDENED STEELS			CAST IRON	ALUMINUM ALLOYS	GRAPHITE	COPPER	PLASTICS	GLASS FILLED PLASTICS	TITANIUM ALLOYS	HEAT RESISTANT ALLOYS	CEMENTED CARBIDE	HARD BRITTLE (NON-METALLIC) MATERIALS
			~55HRC	~60HRC	~70HRC										
◯	◯	◯	◯	◯		◯			◯			◯	◯		



The shank taper angle shown is not an exact value and to avoid contact with the work piece, we recommend the user controls the precise value of this angle. Shank taper angle should not make contact with the work piece.

Total 15 models

Unit (mm)

Model Number	Outside Diameter ϕD	Length of Cut ℓ	Shank Taper Angle Bta	Overall Length L	Shank Diameter ϕd	Price ¥
HMES SP 4010-0250	1	2.5	16°	45	4	6,700
HMES SP 4015-0375	1.5	3.75		45	4	6,700
HMES SP 4020-0500	2	5		45	4	4,100
HMES SP 4025-1000	2.5	10		50	4	6,000
HMES SP 4030-0750	3	7.5		45	6	4,410
HMES SP 4030-1200		12		50	6	6,510
HMES SP 4040-1000	4	10		50	6	4,730
HMES SP 4040-1600		16		60	6	6,830
HMES SP 4060-1500	6	15	—	50	6	5,150
HMES SP 4060-2400		24		60	6	7,880
HMES SP 4080-2000	8	20	—	60	8	8,690
HMES SP 4080-3200		32		80	8	15,400
HMES SP 4100-2500	10	25	—	70	10	11,660
HMES SP 4100-4000		40		90	10	18,150
HMES SP 4120-3000	12	30	—	75	12	14,520

Features 1

40 degree helix angle improves the machined surface quality on side milling.

New groove design offers excellent chip evacuation.

Conventional Type



Helix Angle **30°**



NEW



Helix Angle **40°**

Material : S136 (52HRC)

No cutting mark with mirror surface finishing.

Features 2

Newly designed end cutting edge greatly improves the chip evacuation and bottom surface roughness.

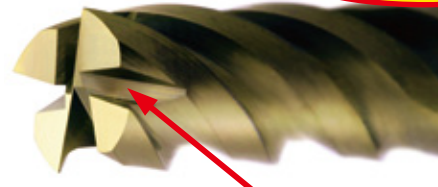
Conventional Type



Conventional design



NEW



Improved pocket design
→Excellent chip evacuation.

Smooth cutting surface at the corner edge.

Material : S136 (52HRC)

◆ Milling Example

- Material : S136 (52HRC)
- Tool Size : $\phi 8 \times 20$
- Coolant : Air Blow

Spindle Speed	Feed Rate	Axial Depth a_p	Radial Depth a_e	Cycle time
5,000 min ⁻¹	1,000 mm/min	12 mm	0.08 mm	1h

Milling Conditions for HMES SP (4Flutes)

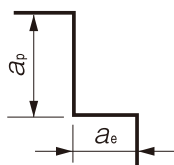
WORK MATERIAL		CARBON STEELS S45C/S50C (~225HB)			ALLOY STEELS SK/SCM/SUS (225~325HB)			PREHARDENED STEELS HARDENED STEELS NAK/SKD (30 ~ 45HRC)			HARDENED STEELS SKD11/61/SKT (45 ~ 55HRC)		
Model Number	Outside Diameter (mm)	Spindle Speed (min ⁻¹)	Velocity (m/min)	Feed Rate (mm/min)	Spindle Speed (min ⁻¹)	Velocity (m/min)	Feed Rate (mm/min)	Spindle Speed (min ⁻¹)	Velocity (m/min)	Feed Rate (mm/min)	Spindle Speed (min ⁻¹)	Velocity (m/min)	Feed Rate (mm/min)
4010	1	20,000	(65)	240	15,000	(45)	215	11,000	(35)	85	7,100	(20)	40
4015	1.5	13,500		245	10,000		215	8,000		90	5,100		50
4020	2	11,000	(70~85)	245	8,500	(55~65)	215	6,400	(40~45)	90	4,000	(25~30)	55
4025	2.5	8,800		370	7,000		245	5,000		90	3,200		55
4030	3	7,400		370	6,400		260	4,500		105	2,800		65
4040	4	5,900		435	5,000		340	3,500		120	2,150		70
4050	5	5,300		590	4,200		415	2,950		120	1,850		75
4060	6	4,400		580	3,500		415	2,450		130	1,500		70
4080	8	3,300		550	2,600		415	1,850		125	1,200		70
4100	10	2,600		525	2,100		405	1,450		125	950		65
4120	12	2,200		525	1,750		405	1,200		120	800		60

◆High speed milling

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Model Number	Outside Diameter (mm)	Spindle Speed (min ⁻¹)	Velocity (m/min)	Feed Rate (mm/min)	Spindle Speed (min ⁻¹)	Velocity (m/min)	Feed Rate (mm/min)	Spindle Speed (min ⁻¹)	Velocity (m/min)	Feed Rate (mm/min)	Spindle Speed (min ⁻¹)	Velocity (m/min)	Feed Rate (mm/min)
4030	3	30,000	(300)	1,500	26,500	(250)	1,075	21,200	(200)	495	15,800	(150)	365
4040	4	23,800		1,755	19,800		1,345	15,800		540	11,900		385
4050	5	19,000		2,115	15,800		1,560	12,700		515	9,500		385
4060	6	15,900		2,095	13,200		1,565	10,600		560	7,900		370
4080	8	11,900		1,985	9,900		1,580	7,900		535	5,900		345
4100	10	9,500		1,920	7,900		1,525	6,300		545	4,700		320
4120	12	7,900		1,885	6,600		1,525	5,300		530	3,900		295

Milling amount for side milling (mm)

Length of Cut Material	2.5D or below	Below 3D	3D or above
45HRC or below	$a_e: 0.07D$ $a_p: 2D$	$a_e: 0.05D$ $a_p: 2.5D$	$a_e: 0.02D$ $a_p: 3.5D$
45HRC or above	$a_e: 0.03D$ $a_p: 1.5D$	$a_e: 0.02D$ $a_p: 2D$	$a_e: 0.01D$ $a_p: 3D$



D : Outside Diameter (mm)

Ex.) 2D or below : Flute Length = Diameter × 2 or below

a_p : Axial Depth (mm)

a_e : Radial Depth (mm)

Note :

- Recommend water soluble or oil coolant.
- Recommend oil coolant for Titanium Alloys and Heat Resistant Alloys.



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Price & Specifications are subject to change without notice.

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